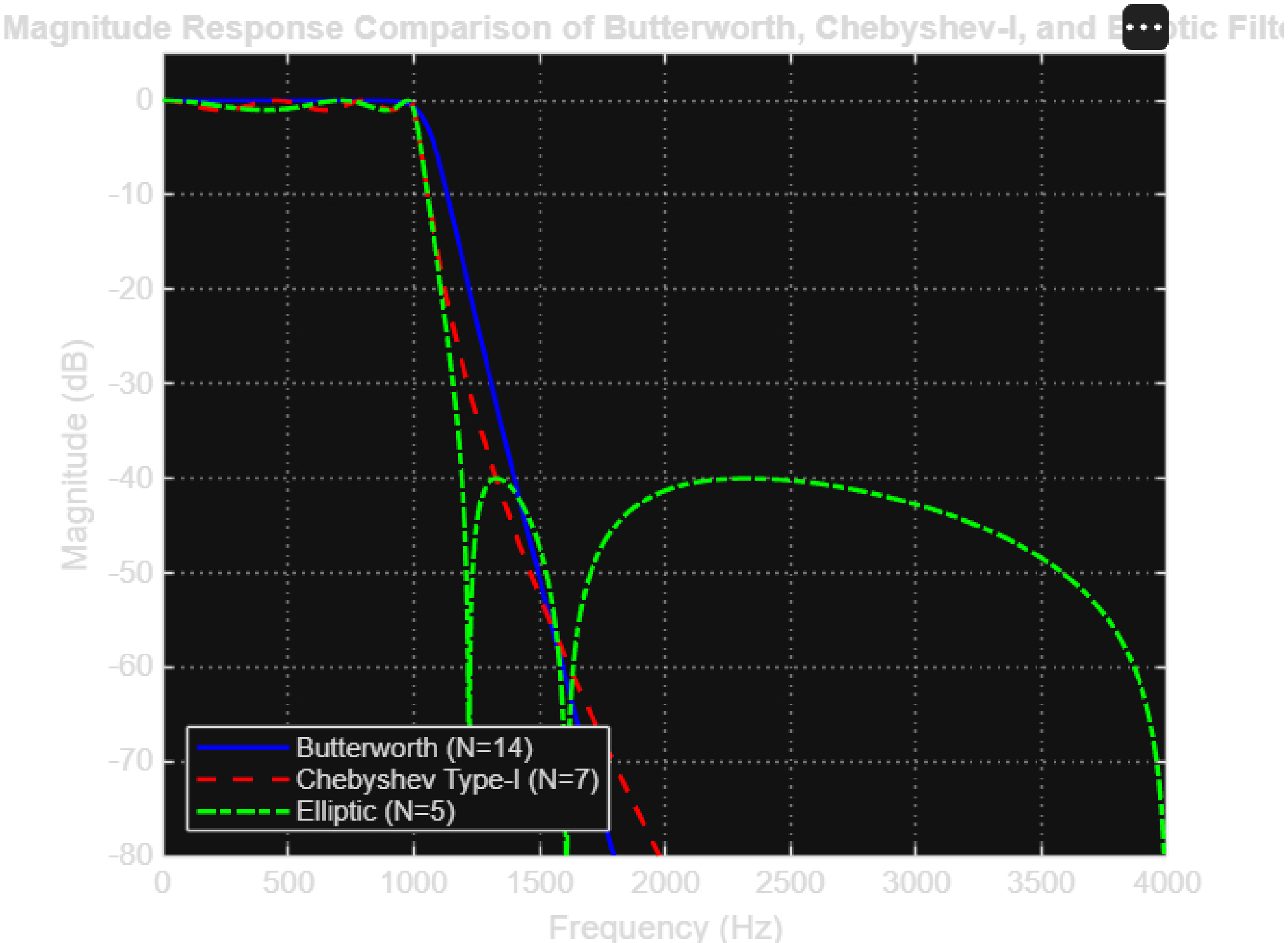
```
Fs = 8000;
Fpass = 1000;
Fstop = 1400;
Rp = 1;
Rs = 40;
Wp = Fpass / (Fs/2);
Ws = Fstop / (Fs/2);
[N_butt, Wn_butt]   = buttord(Wp, Ws, Rp, Rs);
[N_cheb1, Wn_cheb1] = cheb1ord(Wp, Ws, Rp, Rs);
[N_ellip, Wn_ellip] = ellipord(Wp, Ws, Rp, Rs);
[B_butt, A_butt]   = butter(N_butt, Wn_butt, 'low');
[B_cheb1, A_cheb1] = cheby1(N_cheb1, Rp, Wn_cheb1, 'low');
[B_ellip, A_ellip] = ellip(N_ellip, Rp, Rs, Wn_ellip, 'low');
NumPoints = 2048;
[H_butt, f]   = freqz(B_butt, A_butt, NumPoints, Fs);
[H_cheb1, ~]  = freqz(B_cheb1, A_cheb1, NumPoints, Fs);
[H_ellip, ~]  = freqz(B_ellip, A_ellip, NumPoints, Fs);
Mag_butt  = 20 * log10(abs(H_butt));
Mag_cheb1 = 20 * log10(abs(H_cheb1));
Mag_ellip = 20 * log10(abs(H_ellip));
figure('Color', [1 1 1]);
plot(f, Mag_butt, 'b-', 'LineWidth', 1.5); hold on;
plot(f, Mag_cheb1, 'r--', 'LineWidth', 1.5);
plot(f, Mag_ellip, 'g-.', 'LineWidth', 1.5);
grid on;
ax = gca; ax.GridLineStyle = ':'; ax.GridAlpha = 0.4;
title('Magnitude Response Comparison of Butterworth, Chebyshev-I, and Elliptic Filters');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 5]);
```

```
xlim([0 Fs/2]);
legend(sprintf('Butterworth (N=%d)', N_butt), ...
    sprintf('Chebyshev Type-I (N=%d)', N_cheb1), ...
    sprintf('Elliptic (N=%d)', N_ellip), ...
    'Location', 'southwest');
fprintf('============= EXPERIMENT 4 FILTER METRICS =============\n');
fprintf('Specification Details: Fpass=%dHz, Fstop=%dHz, Rp=%ddB, Rs=%ddB\n\n', Fpass, Fstop, Rp, Rs);
fprintf('Filter Type    | Calculated Order (N) | Transition Bandwidth (Hz)\n');
fprintf('-------------------------------------------------------------------\n');
fprintf('Butterworth    |          %d          |           %d\n', N_butt, (Fstop - Fpass));
fprintf('Chebyshev-I    |          %d          |           %d\n', N_cheb1, (Fstop - Fpass));
fprintf('Elliptic       |          %d          |           %d\n', N_ellip, (Fstop - Fpass));
fprintf('===================================================================\n');

```


Magnitude Response Comparison of Butterworth, Chebyshev-I, and Elliptic Filters
Magnitude (dB)
Frequency (Hz)
0
-10
-20
-30
-40
-50
-60
-70
-80
0
500
1000
1500
2000
2500
3000
3500
4000
Butterworth (N=14)
Chebyshev Type-I (N=7)
Elliptic (N=5)